

Mission Opportunity–Constraint Architecture (MOCA): an AI-executable technical management framework for social enterprises

Equations, dynamic policy paths, binding, readout audit, pseudocode, and DAG / knowledge-graph specification

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Abstract. This technical paper consolidates a sequence of practitioner-derived and formally disciplined social-enterprise models into an executable management architecture. The anchor lineage is Dual Costs, Translated Trust Dynamics (TTD), Positional Governance, and the Readout Condition. The resulting *Mission Opportunity–Constraint Architecture* (MOCA) represents mission not as a single objective or organizational “type” but as a dynamic configuration that can (i) create economic opportunities, (ii) constrain otherwise admissible choices, and (iii) alter the ranking of admissible choices through a mission objective. The framework separates mission-attributable economic sacrifice from binding expenditure, direct revenue forgone, stakeholder incidence, social outcome, and epistemic identification. It extends one-period choice into a policy-path model with liquidity, receivables, inventory, debt, managerial and coordination capacity, operational and symbolic trust, organizational topology, mission-obligation stock, ecosystem feedback, uncertainty, and portability. A readout layer requires every calculated claim to expose its identification set and minimal sufficient support bundle; unidentified values are returned as NI, not zero. The paper provides a complete equation inventory, pseudocode, an AI calculation contract, two acyclic dependency graphs, a knowledge-graph schema, a worked example, and a falsification protocol with leakage control. Development-stage real-enterprise batteries are treated only as finite diagnostics, not population validation. The objective is reproducible calculation and falsifiable management diagnosis rather than a universal law of social enterprise.

Keywords. social enterprise; constrained optimization; dynamic programming; hybrid organizations; mission drift; trust; governance; partial identification; provenance; knowledge graph; AI-executable management; policy design.

1. PURPOSE AND SCOPE

Social enterprises combine commercial survival with social commitments that can change which transactions, products, workers, suppliers, capital providers, prices, data practices, or allocations are admissible. The literature on hybridity and social entrepreneurship documents multiple logics, resource tensions, and value-creation / value-capture trade-offs [21, 2, 7]. The present paper does not replace those literatures. It specifies a *calculation architecture* for a narrower operational question: given a declared mission, organizational state, feasible actions, evidence base, and time horizon, what choices are admissible, what private economic opportunities and sacrifices are mission-attributable, what binding mechanisms hold the mission, whether the enterprise remains viable, and which of those conclusions are actually identified by the available readouts?

The framework is anchored in four preceding constructions. *Dual Costs* separates production cost from the cost of maintaining legitimacy and demonstrable trust [12]. *TTD* separates operational from symbolic trust and models survival under margins, action lag, hazard, and growth cost [13]. *Positional Governance* isolates a class of enterprises that hold a trans-

action and must sometimes refuse completion despite being paid to complete it; it also distinguishes binding expenditure from the revenue forgone when a condition is actually imposed [14]. The *Readout Condition* requires claim-level distinctions to have an adequate provenance path, licensed attribution, and recoverable augmentation [15].

1.1. Claim tiers and the formal boundary

MOCA is a technical synthesis, not a theorem that all social enterprises obey. The accompanying Universal Solver repository contains generic Coq artifacts showing, among other things, that a record can be invariant to admissible relabeling while still being strictly lossy; MOCA imports the discipline of not equating readout with latent state, not a machine proof of social-enterprise economics [16]. We therefore distinguish:

- **Th-coqc:** axiom-free machine-checked generic structural claims in the formal repository;
- **standard-math:** established optimization, game-theoretic, or dynamic-programming constructions used here [3, 11, 22, 5];

- **model:** MOCA definitions and identities conditional on declared inputs;
- **finite-diagnostic:** retrospective or prediction-frozen case batteries;
- **NI/Open:** distinctions not identified by the current evidence or bridge assumptions.

Finite batteries must never be promoted to “all enterprises” claims.

2. CORE DATA MODEL

2.1. Reality, readout, and non-equivalence

Let E_t denote the latent enterprise state relevant to question Q . A source or institutional channel a provides a readout

$$R_{a,t} = K_a \theta(E_t) + \eta_{a,t}^{\text{sel}} + \eta_{a,t}^{\text{map}} + \eta_{a,t}^{\text{self}}. \quad (1)$$

The only universal discipline used operationally is

$$R_{a,t} \neq E_t \quad \text{in general.} \quad (2)$$

Accounting records, audit files, impact reports, customer data, or interviews are therefore typed as *readouts*, not as the enterprise reality itself.

2.2. Enterprise state

TTD originally used cash, operational trust, symbolic trust, hazard memory, and regime. Stress tests across financing, PAYGo, platform, and multi-entity cases require an expanded state:

$$X_t = (\text{Liq}_t, AR_t, \text{Inv}_t, \text{Debt}_t, \text{Cap}_t^{\text{mgr}}, \text{Cap}_t^{\text{coord}}, T_t^{\text{op}}, T_t^{\text{sym}}, H_t, \sigma_t). \quad (3)$$

Here $\text{Cap}_t^{\text{mgr}}$ is managerial bandwidth and $\text{Cap}_t^{\text{coord}}$ is the capacity to coordinate essential actors. Symbolic trust is stakeholder-specific:

$$T_t^{\text{sym}} = (T_{\text{customer}}^{\text{sym}}, T_{\text{worker}}^{\text{sym}}, T_{\text{supplier}}^{\text{sym}}, T_{\text{regulator}}^{\text{sym}}, T_{\text{funder}}^{\text{sym}}, T_{\text{investor}}^{\text{sym}}, \dots)_t. \quad (4)$$

The complete management state is

$$S_t = (X_t, Z_t, O_t, \Omega_t^M, Q_t^M, W_t), \quad (5)$$

where Z_t is the ecosystem, O_t the organizational topology, Ω_t^M the mission-enabled opportunity set, Q_t^M the outstanding mission-obligation stock, and W_t the binding architecture.

2.3. Organizational topology

A social enterprise can be distributed across legal entities, suppliers, funders, franchises, cooperatives, or protected-party relationships. Define

$$O_t = (N_t, \mathcal{E}_t, \Gamma_t), \quad (6)$$

where nodes N_t are organizational actors, edges $\mathcal{E}_t$ are resource / contract / control relations, and Γ_t stores governance rights. A mission rule may reside on a node *or* on an edge.

3. MISSION ARCHITECTURE

3.1. Mission-binding vector

Exclusive enterprise “types” are retained only as coarse projections. The operational representation is a question-relative vector

$$b_{Q,t} = (b_A, b_L, b_S, b_P, b_{\text{Pay}}, b_T, b_D, b_K, b_R, \dots)_t \quad (7)$$

for Allocation, Labour, Sourcing, Product / Offering, Payment / Customer Finance, Transaction, Data / Information, Capital / Governance, and Process / Lifecycle. This vector is an *audit coordinate*, not a claim that nature contains exactly these dimensions.

3.2. Locus-specific binding

For each locus j , define a binding architecture

$$W_{j,t} = (\text{Rule, Record, Rights, Voice, Control, Audit, Capital})_{j,t}. \quad (8)$$

Only a constitutive subset R_j^* is required at a given locus. Let $b_{jr} \in \{0, 1\}$ indicate whether constitutive requirement r is satisfied. Then

$$q_j = \prod_{r \in R_j^*} b_{jr}, \quad \hat{\beta}_j = q_j a_j, \quad (9)$$

where $a_j \in [0, 1]$ is assurance / readout strength. If any constitutive element fails, $q_j = 0$. This is deliberately *not* a universal latent “binding score.” Positional Governance provides a stricter transaction-gate instance in which declaration, protected-party rights, separate condition records, conflict disclosure, external audit, and capital admissibility are load-bearing [14, 20].

3.3. Mission as opportunity, constraint, and objective

The master decomposition is

$$\begin{aligned} \text{Mission} &= \text{Opportunity Creation} \\ &+ \text{Hard Constraint} + \text{Mission Objective.} \end{aligned} \quad (10)$$

This is a representational decomposition, not an empirical claim that all three terms are nonzero in every enterprise. Mission opportunities are split into exogenous and endogenous components:

$$\Omega_t^M = \Omega_t^{\text{exo}} \cup \Omega_t^{\text{endo}}. \quad (11)$$

Because mission credibility can itself create network access, premium demand, supplier capability, or institutional contracts, the endogenous component may satisfy a fixed-point relation

$$\Omega_t^{\text{endo}} = G(\Omega_t^{\text{endo}}, b_t, W_t, X_t, Z_t, O_t). \quad (12)$$

If multiple fixed points exist, Ω_t^{endo} and downstream economic values are *set-identified* until an equilibrium-selection rule is independently justified.

4. STATIC ECONOMIC ARCHITECTURE

4.1. Feasible set and viability gate

Let $\mathcal{F}_t^0$ be the mission-relaxed conventional feasible set and define

$$\widetilde{\mathcal{F}}_t = \mathcal{F}_t^0 \cup \Omega_t^M. \quad (13)$$

Each hard mission constraint is

$$G_{j,t}(x, X_t, Z_t, O_t, H_t, \omega_t) \geq \tau_{j,t}^{\text{auth}}, \quad (14)$$

where H_t is history, ω_t an uncertain state, and τ^{auth} the authorized threshold. The mission-admissible set is

$$\mathcal{F}_t^{SE} = \{x \in \widetilde{\mathcal{F}}_t : G_{j,t}(x, \dots) \geq \tau_{j,t}^{\text{auth}} \quad \forall j\}. \quad (15)$$

Before optimization, compute

$$v_t = \mathbf{1}[\mathcal{F}_t^{SE} \neq \emptyset]. \quad (16)$$

If $v_t = 0$, no economic optimum is reported; the diagnosis is *mission infeasibility under the current state*, not mission drift.

4.2. Mission objective and four benchmarks

Let private economic value be $V_t(x)$ and mission ranking be $\Psi_t^M(x)$. The mission decision objective is

$$J_t(x) = V_t(x) + \Psi_t^M(x). \quad (17)$$

When $v_t = 1$, define four benchmarks:

$$x_t^0 \in \arg \max_{x \in \mathcal{F}_t^0} V_t(x), \quad (18)$$

$$x_t^\Omega \in \arg \max_{x \in \widetilde{\mathcal{F}}_t} V_t(x), \quad (19)$$

$$x_t^G \in \arg \max_{x \in \mathcal{F}_t^{SE}} V_t(x), \quad (20)$$

$$x_t^{SE} \in \arg \max_{x \in \mathcal{F}_t^{SE}} [V_t(x) + \Psi_t^M(x)]. \quad (21)$$

Set-valued $\arg \max$ is intentional: existence and uniqueness are separate proof or numerical obligations.

4.3. Opportunity value and mission sacrifice

Define mission opportunity value

$$O_{M,t} = V_t(x_t^\Omega) - V_t(x_t^0) \geq 0, \quad (22)$$

hard-constraint sacrifice

$$C_{G,t} = V_t(x_t^\Omega) - V_t(x_t^G) \geq 0, \quad (23)$$

and mission-objective sacrifice

$$C_{\Psi,t} = V_t(x_t^G) - V_t(x_t^{SE}) \geq 0. \quad (24)$$

Total mission-attributable private economic sacrifice is

$$C_{M,t} = C_{G,t} + C_{\Psi,t}. \quad (25)$$

The central economic identity is

$$V_t(x_t^{SE}) - V_t(x_t^0) = O_{M,t} - C_{G,t} - C_{\Psi,t} = O_{M,t} - C_{M,t}. \quad (26)$$

This equation separates mission-created opportunity from mission-imposed sacrifice. It does not aggregate social welfare.

4.4. Direct revenue forgone is not total sacrifice

For position-holding or other refusal-gate cases, direct revenue forgone remains a useful operational readout:

$$RF_{j,t} = Rev_{j,t}^{\text{counterfactual}} - Rev_{j,t}^{\text{observed}}. \quad (27)$$

In general,

$$RF_{j,t} \neq C_{G,j,t}, \quad (28)$$

because refusal may create reputation, lower future risk, or open institutional opportunities.

4.5. Constraint interaction and Shapley attribution

Mission constraints can interact. Let V^S be the best private economic value when exactly the constraint subset $S \subseteq J$ is active on the mission-expanded opportunity set. Define

$$C(S) = V^\Omega - V^S. \quad (29)$$

Pairwise interaction is

$$\Gamma_{jk} = C(\{j, k\}) - C(\{j\}) - C(\{k\}), \quad (30)$$

and the Shapley attribution is [22]

$$\phi_j = \sum_{S \subseteq J \setminus \{j\}} \frac{|S|! (|J| - |S| - 1)!}{|J|!} \times [C(S \cup \{j\}) - C(S)]. \quad (31)$$

When the coalition value is fully specified, $\sum_j \phi_j = C(J)$. Empirically, use $\hat{\phi}_j = \phi_j(\hat{C}, B, A_k)$: attribution is model-relative, not a natural decomposition supplied by the world.

4.6. Local shadow price under differentiability

For a differentiable continuous problem with $g_j(x) \geq \tau_j$, the Lagrangian can be written

$$\mathcal{L}(x, \lambda) = V(x) + \sum_j \lambda_j [g_j(x) - \tau_j], \quad \lambda_j \geq 0. \quad (32)$$

Under standard regularity conditions, complementary slackness gives

$$\lambda_j [g_j(x^*) - \tau_j] = 0, \quad (33)$$

and the local marginal private value of tightening threshold j is

$$-\frac{\partial V^*}{\partial \tau_j} = \lambda_j. \quad (34)$$

This is a standard KKT interpretation [11, 5]; it should not be confused with the global, interaction-aware Shapley attribution.

5. COST, ACCOUNTING, AND TTD PLUG-IN

5.1. Binding expenditure versus economic sacrifice

Dual Costs and Positional Governance motivate a strict separation:

$$C_{M,t} \neq C_t^{\text{bind}}. \quad (35)$$

C_M is counterfactual private economic value sacrificed by mission; C^{bind} is cash or resource expenditure required to make the mission credible, recorded, auditable, or enforceable. The latter can include certification, impact evaluation, complaints systems, protected-party rights, audit, or governance covenants [12, 14, 23].

5.2. Cash ledger

To avoid double counting, counterfactual sacrifice is *not* automatically subtracted from cash flow. A management ledger can be written

$$\text{Liq}_{t+1} - \text{Liq}_t = CF_t^{\text{op}} + U_t^{\text{cash}} - C_t^{\text{bind}} - I_t^{\text{MC}} - DS_t - \Delta WC_t - \text{Loss}_t. \quad (36)$$

Here I^{MC} is mission-enabled market / capability-creation investment, DS debt service, and ΔWC working-capital change. The accounting audit residual is

$$R_T^{\text{cash}} = (\text{Liq}_T - \text{Liq}_0) - \sum_{t < T} (\text{Liq}_{t+1} - \text{Liq}_t), \quad (37)$$

which must equal zero for an internally closed discrete ledger.

5.3. TTD revenue and trust dynamics

TTD v3 supplies an optional dynamic plug-in [13]. In its original scalar symbolic-trust form,

$$\text{Rev}_t = r_0 [0.6 + 0.8 \tanh(T_t^{\text{op}})] \times (1 + \beta T_t^{\text{sym}})(1 + \sigma_r \epsilon_t), \quad (38)$$

$$G_t = \max(0, T_t^{\text{sym}} - \kappa T_t^{\text{op}}), \quad (39)$$

$$p_t^{\text{haz}} = \text{clip}[h_0(1 + h_m H_t) e^{-h_p T_t^{\text{op}}} + \phi G_t, 0, 0.95], \quad (40)$$

$$\text{Ovh}_t = \max(\text{Ovh}_t^{\text{floor}}, \text{Ovh}_0[1 - \alpha_{\text{eff}} T_t^{\text{op}}]) + \gamma \max(0, \text{Liq}_t - \text{Liq}^{\text{scale}}), \quad (41)$$

$$T_{t+1}^{\text{op}} = T_t^{\text{op}}(1 - \delta_{\text{op}}), \quad (42)$$

$$T_{t+1}^{\text{sym}} = T_t^{\text{sym}}(1 - \delta_{\text{sym}}). \quad (43)$$

Action cash effects can be immediate while trust / hazard effects arrive after lag ℓ . The original TTD cash update is

$$\text{Liq}_{t+1} = \text{Liq}_t + \text{Rev}_t - \text{Ovh}_t - \text{hit}_t + \Delta \text{Liq}(a_t), \quad (44)$$

with hard survival constraints

$$\text{Liq}_t \geq \text{Liq}^{\text{min}}, \quad T_t^{\text{op}} \geq T_{\text{min}}^{\text{op}}, \quad H_t \leq H^{\text{max}}. \quad (45)$$

When the vector trust representation in (4) is used, each stakeholder-specific expectation gap is

$$G_{a,t} = \max(0, T_{a,t}^{\text{sym}} - \kappa_a T_{a,t}^{\text{op}}). \quad (46)$$

Any projection of the trust vector back to a scalar revenue or hazard term must declare its weights; the projection is a decision / model augmentation, not a direct readout.

6. DYNAMIC POLICY-PATH ARCHITECTURE

6.1. Mission-obligation stock

Some actions create future mission liabilities: warranties, aftercare, training, affordability covenants, take-back, referral, or long-term supplier support. Define

$$Q_{t+1}^M = (1 - \delta_Q) Q_t^M + q^M(x_t) - \text{Fulfilled}_t. \quad (47)$$

A firm can therefore be profitable at t while accumulating mission obligations that destroy future viability.

6.2. Ecosystem feedback

The ecosystem is not purely exogenous:

$$Z_{t+1} = \mathcal{Z}(Z_t, x_t, O_t, u_t, \epsilon_t). \quad (48)$$

Enterprises can create distribution infrastructure, alter supplier capabilities, influence norms, or induce policy responses.

6.3. State transition and dynamic viability

The complete transition is

$$S_{t+1} = \mathcal{T}(S_t, x_t, u_t, Q_t^M, \epsilon_t). \quad (49)$$

Dynamic viability requires

$$S_t \in \mathcal{V} \quad \forall t \leq T, \quad (50)$$

and a survival probability can be reported as

$$P_{\text{surv}}(T) = \mathbb{P}(S_t \in \mathcal{V} \quad \forall t \leq T). \quad (51)$$

6.4. Policy path and Bellman form

Let $\pi : S_t \mapsto x_t$ be a management policy and let Π^{SE} be the set of policies that satisfy mission constraints along the path. The dynamic management objective is

$$W^{\text{SE}}(S_t) = \sup_{\pi \in \Pi^{\text{SE}}} \mathbb{E} \left[\sum_{k=t}^T \delta^{k-t} \left(V_k + \Psi_k^M + \Psi_k^u - C_k^{\text{bind}} - I_k^{\text{MC}} \right) \right]. \quad (52)$$

This is a standard multi-stage decision form [3]. Ψ^u records incentive effects of a policy instrument; policy support is not assumed to be an additive cash transfer only.

6.5. Uncertainty

For stochastic constraints, the model must declare its risk rule. A chance constraint is

$$\mathbb{P}[G_{j,t}(x_t, \omega_t) \geq 0 \mid \mathcal{I}_t] \geq 1 - \alpha_j, \quad (53)$$

while a robust alternative is

$$\inf_{\omega \in \mathcal{U}_t} G_{j,t}(x_t, \omega) \geq 0. \quad (54)$$

The tolerance α_j and uncertainty set $\mathcal{U}_t$ are policy / model choices and therefore belong in provenance.

7. GOVERNANCE, INTEGRITY, DRIFT, AND STAKEHOLDERS

7.1. Conflict and mission integrity

Let

$$\chi_{j,k} = \mathbf{1}[\text{identifiable conflict at } (j, k)], \quad (55)$$

$$h_{j,k} = \mathbf{1}[\text{mission honored at } (j, k)]. \quad (56)$$

Mission integrity is defined *only* when conflict opportunities are identifiable:

$$I_j = \frac{\sum_k \chi_{j,k} h_{j,k}}{\sum_k \chi_{j,k}} \quad \text{only if ID}(\chi_j) = 1. \quad (57)$$

If the denominator is zero or records cannot distinguish “no conflict” from “conflict not detected,” return $I_j = \text{NI}$, not 0 or 1. Positional Governance’s zero-invocation warning is a special case: a gate that never fails may be a *non-readout* rather than evidence of safety [14].

7.2. Authorized revision and mission drift

Let the mission threshold be state-contingent under a pre-authorized rule γ_j :

$$\tau_{j,t}^{\text{auth}} = \tau_j(S_t; \gamma_j). \quad (58)$$

Then a hard-constraint drift indicator is

$$\text{Drift}_{j,t} = \mathbf{1}[\tau_{j,t}^{\text{obs}} < \tau_{j,t}^{\text{auth}}(S_t)] \chi_{j,t}. \quad (59)$$

Changing an implementation mechanism or invoking a declared hardship rule is therefore not automatically mission drift.

7.3. Stakeholder incidence and social outcome

Private economic value at the firm level is not social welfare. For stakeholder g , a distributional ledger may be written

$$\Delta V_{g,t} = O_{g,t} - C_{G,g,t} - C_{\Psi,g,t} - C_{g,t}^{\text{bind}} + U_{g,t}. \quad (60)$$

Social outcomes are represented as a stakeholder vector

$$Y_t^{\text{soc}} = (Y_{\text{beneficiary}}, Y_{\text{workers}}, Y_{\text{suppliers}}, Y_{\text{local market}}, Y_{\text{community}}, Y_{\text{environment}}, \dots)_t. \quad (61)$$

If a policy analysis aggregates this vector,

$$W_t^{\text{soc}} = \sum_g \omega_g Y_{g,t}, \quad (62)$$

then the normative weights ω_g must be disclosed. Mission integrity, social outcome, and social welfare remain distinct objects.

7.4. Minimum support and portability

Static minimum mission-viability support is

$$U_t^{\min} = \inf\{U \geq 0 : \mathcal{F}_t^{SE}(U) \neq \emptyset\}, \quad (63)$$

while a dynamic version is

$$U_{0:T}^{\min} = \inf\{U : S_t(U) \in \mathcal{V} \ \forall t \leq T\}. \quad (64)$$

For ecosystem transfer $Z_A \rightarrow Z_B$, define a model-relative portability quantity

$$\text{Port}(M; Z_A \rightarrow Z_B) = \mathbb{P}[\mathbf{v}_B = 1 \wedge I_B \geq I_{\min}]. \quad (65)$$

This separates growth in one ecosystem from replication while preserving mission in another.

8. READOUT-BOUNDED INFERENCE

8.1. Identified sets

For any calculated quantity q , let r be the observed records and A_k the declared augmentation layer. The identified set is

$$\mathcal{I}_q(r; A_k) = \{q(e') : e' \in \Gamma_k(r)\}. \quad (66)$$

Point identification requires

$$|\mathcal{I}_q| = 1. \quad (67)$$

If not, output a set, interval, scenario family, or NI. Partial identification follows the conservative logic associated with Manski [17].

8.2. Minimal sufficient support bundles and E-A-D

For claim distinction d , let $P(d) = (V_d, E_d, \tau_d)$ be a typed provenance DAG and let

$$\mathcal{S}(d; y, \kappa) \subseteq 2^{V_d} \quad (68)$$

be the antichain of minimal jointly sufficient support sets under the declared audit context [15, 8, 6, 1]. A represented basis B is *attribution-adequate* only if

$$\exists S \in \mathcal{S}(d; y, \kappa) \text{ such that } S \subseteq B. \quad (69)$$

The Readout Condition is

$$RC(d) = E(d) \wedge A(d) \wedge D(d), \quad (70)$$

where E is provenance existence, A is licensed attribution, and D is disclosure / recoverability of material augmentation.

For defeated provenance nodes $\Delta \subseteq V_d$, surviving supports are

$$\mathcal{S}_\Delta(d) = \{S \in \mathcal{S}(d) : S \cap \Delta = \emptyset\}, \quad (71)$$

and support vulnerability is

$$v_d(\Delta) = \mathbf{1}[\mathcal{S}_\Delta(d) \neq \emptyset]. \quad (72)$$

A system implementing MOCA must downgrade or retract a claim when all adequate support bundles are defeated.

8.3. Readout gaming and Goodhart pressure

Let M_t^* be a latent mission-relevant state and let an organization be able to act strategically on the measurement process:

$$R_t = R(M_t^*, a_t^{\text{game}}, \eta_t). \quad (73)$$

Then

$$R_t \uparrow \nRightarrow M_t^* \uparrow. \quad (74)$$

This is a generic measurement warning related to Goodhart-type target effects [10]; it is not a claim that every KPI is gamed.

9. PREDICTION PROVENANCE AND FALSIFICATION

Retrospective case fit is weak evidence because model construction can absorb outcomes after the fact. A stronger finite-diagnostic protocol freezes prediction rules before outcome retrieval and records leakage.

For prediction d , define

$$Pred(d) = (\hat{y}_d, \mathcal{S}_{\text{pre}}(d), t_{\text{freeze}}, Leak_d). \quad (75)$$

Outcome evidence has its own support family $\mathcal{S}_{\text{out}}(d)$. A scored test exists only when

$$Testable(d) = \mathbf{1}[Leak_d = 0] \mathbf{1}[\mathcal{S}_{\text{out}}(d) \neq \emptyset]. \quad (76)$$

For set-valued prediction and outcome regions,

$$Hit(d) = Testable(d) \mathbf{1}[\hat{Y}_d \cap Y_d^{\text{out}} \neq \emptyset], \quad (77)$$

$$Refute(d) = Testable(d) \mathbf{1}[\hat{Y}_d \cap Y_d^{\text{out}} = \emptyset]. \quad (78)$$

An outcome with inadequate support is NI, not a win for the model.

10. AI-EXECUTABLE CALCULATION CONTRACT

The framework is designed so that a human or AI system can calculate it without silently inventing missing quantities. Table 1 is the minimum contract.

10.1. Required output object

Every computed quantity should be emitted as a structured object:

Object: `QuantityResult`

```
QuantityResult = {
  name: string,
  value: number | interval | set | "NI",
  unit: string | null,
  equation_id: string,
  time_index: int | [start, end],
  identification: "point" | "set" | "NI",
  assumptions: [id...],
  support_bundles: [[source_id...], ...],
  defeated_sources: [source_id...],
  tier: "model" | "standard-math"
      | "finite-diagnostic" | "NI/Open",
  notes: string
}
```

A bare number without this metadata is not a complete MOCA result.

11. PSEUDOCODE

Algorithm 1 — One-period diagnostic

```
function MOCA_DIAGNOSE(data, question_Q):
  validate_schema(data)
  R <- collect_readouts(data.evidence)
  build_provenance_DAG(R)
  X <- assemble_state(data.state)
  0 <- assemble_topology(data.organizations,
                        data.edges,
                        data.control_rights)
  b <- locate_mission_loci(data.mission, question_Q)

  for each locus j in active(b):
    W[j] <- evaluate_binding_components(j)
    q[j] <- product(W[j][r]
                  for r in constitutive[j])
    beta_hat[j] <- q[j] * assurance[j]
    if assurance known else NI

  Omega_exo <- declared_external_opportunities()
  Omega_endo <- solve_or_bound_fixed_point(
    G_omega, b, W, X, Z, 0)
  F_tilde <- union(F0, Omega_exo, Omega_endo)
  F_SE <- {x in F_tilde :
    all mission constraints pass}

  if F_SE is empty:
    U_min <- solve_min_support_for_nonempty(F_SE)
    return INFEASIBLE with U_min and provenance

  x0 <- argmax_set(F0, V)
  x0_omega <- argmax_set(F_tilde, V)
  xG <- argmax_set(F_SE, V)
  xSE <- argmax_set(F_SE, V + Psi_M)

  O_M <- V(x0_omega) - V(x0)
  C_G <- V(x0_omega) - V(xG)
  C_Psi <- V(xG) - V(xSE)
  C_M <- C_G + C_Psi

  if multiple constraints
    and coalition values available:
    phi <- shapley_attribution(coalitions)
  else:
    phi <- NI

  integrity <- compute_integrity_only_if(
    conflicts_identified)
  readout_audit <- EAD_and_identification_audit(
    all_claims)
```

```

return structured_results(
  X, 0, b, W, F_SE, x0, xOmega, xG, xSE,
  O_M, C_G, C_Psi, C_M, phi, integrity,
  readout_audit)

```

Algorithm 2 — Dynamic management simulation

```

function MOCA_SIMULATE(initial_state, policy_pi,
  horizon_T):
  S <- initial_state
  path <- []
  for t in 0..T-1:
    if S not in viability_region:
      return FAILURE(path,
        cause="viability exit")
    actions <- feasible_actions(S)
    x <- policy_pi(S, actions)
    verify_mission_constraints(x, S)
    cash <- cash_ledger_update(S, x)
    trust_hazard <- TTD_update_with_lag(S, x)
    Q_M <- obligation_stock_update(S.Q_M, x)
    Z_next <- ecosystem_update(S.Z, x,
      policy_support)
    S_next <- transition(S, x, cash,
      trust_hazard, Q_M, Z_next)
    path.append(audit_record(t, S, x, S_next))
    S <- S_next
  return PATH_RESULT(path,
    survival = true,
    discounted_value = Bellman_sum(path),
    integrity = path_integrity(path),
    provenance = path_provenance(path))

```

Algorithm 3 — Readout and claim audit

```

function AUDIT_CLAIM(claim_d, evidence_graph,
  assumptions):
  Gamma <- construct_identified_alternatives(
    evidence_graph, assumptions)
  I_q <- evaluate_quantity_over(Gamma)
  support_family <-
    minimal_jointly_sufficient_supports(claim_d)

  if support_family is empty:
    return NI("no adequate support bundle")
  if represented_basis_contains_no_bundle():
    return FAIL_ATTRIBUTION
  if material_augmentation_not_disclosed():
    return SILENT_LIFT
  if cardinality(I_q) == 1:
    return POINT_IDENTIFIED(only_element(I_q))
  else:
    return SET_IDENTIFIED(I_q)

```

Algorithm 4 — Prediction-falsification protocol

```

function SCORE_PREDICTION(prediction,
  outcome_evidence):
  if prediction.leak == 1:
    return EXCLUDED("leakage before freeze")
  S_out <- adequate_outcome_supports(
    outcome_evidence)
  if S_out is empty:
    return NI("outcome not identified")
  Y_out <- identified_outcome_set(
    outcome_evidence)
  if intersection(prediction.Y_hat, Y_out)
    is empty:
    return REFUTED
  else:
    return NOT_REFUTED

```

12. EXECUTABLE DAG AND KNOWLEDGE-GRAPH REPRESENTATION

A single omnibus graph is unsafe here because three different relations are easy to conflate: computational dependency, dynamic state transition, and epistemic provenance. MOCA therefore uses separate directed acyclic graphs (DAGs). Dynamic feedback is made acyclic by *time-unrolling*: within an epoch, information flows from typed inputs to a decision and then to updates; recurrence appears only as an edge into the next state, S_{t+1} . The provenance graph separately records which readouts, models, assumptions, and support bundles license a claim. Neither graph is asserted to be a causal DAG unless causal semantics are independently supplied.

12.1. Management computation DAG

Figure 1 gives the minimum executable dependency order for one decision epoch. It is deliberately coarser than the full knowledge graph: the purpose is to make calculation order and missing-input propagation obvious to a human or AI implementation.

12.2. Prediction provenance and falsification DAG

Prediction testing has a different dependency structure and must not share arrows with the management graph. Figure 2 makes the separation explicit. Pre-outcome evidence must be frozen before outcome retrieval; outcome evidence has its own adequate support family; leakage excludes a prediction from scoring; and later defeaters can downgrade a prior verdict by destroying all surviving adequate support bundles.

12.3. Machine-readable knowledge graph

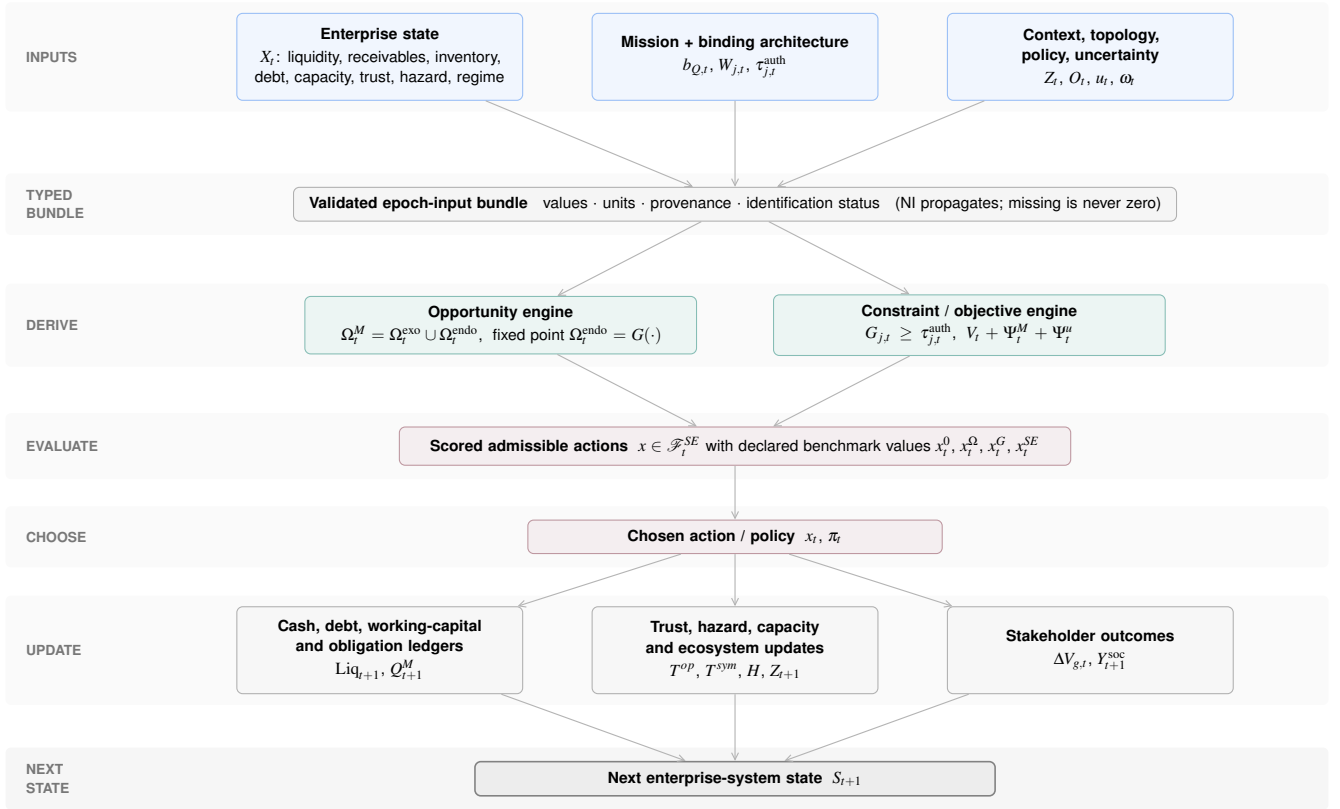
The accompanying JSON-LD file serializes the same typed entities and relations without imposing page-layout semantics. Implementations should preserve the distinction between (i) management-computation dependencies, (ii) time-indexed state transitions, and (iii) provenance / support dependencies. A provenance DAG may be embedded in a larger knowledge graph, but cycles in the larger graph must never be interpreted as cycles in a single claim's support structure.

13. WORKED NUMERICAL EXAMPLE

This example is synthetic and is included only to demonstrate calculation order. Monetary units are arbitrary. Assume conventional actions $\mathcal{F}^0 = \{a_0, a_1, a_3\}$, and mission credibility creates an additional institutional-contract action a_2 . Private values and mission scores are:

Action	V	Ψ^M	Pass?
a_0 : premium market	120	0	no
a_1 : low-income direct	95	40	yes
a_2 : institutional contract	132	25	yes
a_3 : supplier-upgrading	110	50	yes

Then $x^0 = a_0$, $x^\Omega = a_2$, $x^G = a_2$, while $V + \Psi^M$ selects



Recurrence is time-unrolled: S_{t+1} re-enters as S_t of the next epoch, so no arrow returns within the figure.

Figure 1: MOCA management-computation DAG for one time-unrolled decision epoch. All arrows are computational dependencies. Dynamic recurrence is represented only through $S_t \rightarrow x_t \rightarrow S_{t+1}$, so the figure is acyclic. The graph does not by itself assert causal identification.

$x^{\text{SE}} = a_3$. Therefore

$$O_M = 132 - 120 = 12, \quad (79)$$

$$C_G = 132 - 132 = 0, \quad (80)$$

$$C_\Psi = 132 - 110 = 22, \quad (81)$$

$$\Delta V^{\text{SE}} = 12 - 0 - 22 = -10, \quad (82)$$

which equals $V(a_3) - V(a_0) = 110 - 120 = -10$, verifying Eq. (26).

Suppose $\text{Liq}_t = 300$, operating cash flow is 110, cash support is 20, binding expenditure is 8, market-creation investment is 12, debt service is 5, and working capital rises by 15. Equation (36) gives

$$\text{Liq}_{t+1} = 300 + 110 + 20 - 8 - 12 - 5 - 15 = 390. \quad (83)$$

Notice that $C_M = 22$ is *not* subtracted again from liquidity.

For the TTD plug-in, take $r_0 = 250$, $T^{\text{op}} = 0.55$, $T^{\text{sym}} = 0.75$, $\beta = 0.10$, and no revenue shock. Equation (38) gives $\text{Rev} \approx 268.86$. With $\kappa = 1.2$, Eq. (39) gives $G = 0.09$. If $h_0 = 0.05$, $h_m = 0.9$, $H = 0.4$, $h_p = 0.75$, and $\phi = 0.07$, then

Eq. (40) gives $p^{\text{haz}} \approx 0.0513$. These numbers are illustrative, not calibrated.

Finally, if the source for the claim “ a_2 exists because of mission credibility” is only a founder statement and no contract, counterfactual, or independent buyer evidence distinguishes that attribution, the AI must return O_M as set-identified or NI, even though the arithmetic above is internally correct. *Arithmetic correctness does not repair attribution failure.*

14. RECOMMENDED NUMERICAL METHODS

- **Finite discrete actions:** enumerate all actions, filter by constraints, and compute the four benchmarks exactly.
- **Convex continuous problems:** solve the constrained optimization numerically and use KKT multipliers for local sensitivity [5].
- **Mixed-integer governance rules:** formulate binary binding / eligibility variables in MILP / MINLP form.
- **Stochastic paths:** use scenario trees, Monte Carlo simulation, or dynamic programming; report the scenario model

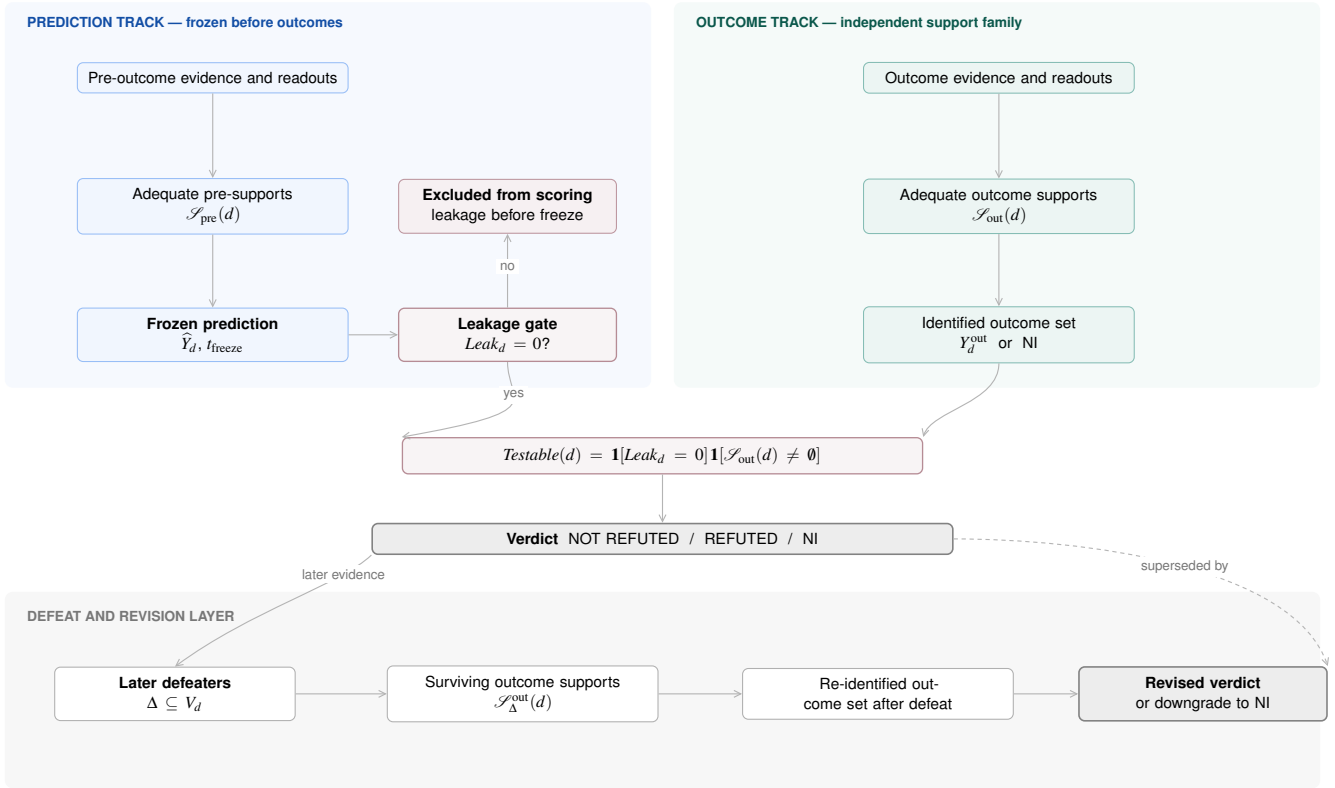


Figure 2: Prediction Provenance and Falsification Module (PPFM) as a separate DAG. Prediction support and outcome support are distinct families. A prediction enters scoring only when the freeze precedes outcome leakage and the outcome is adequately supported. Later node-specific defeaters update the surviving support family and can force a previous verdict back to NI; the dashed edge marks supersession, not re-derivation.

and random seed.

- **Fixed-point opportunities:** solve for all stable fixed points where feasible; if equilibrium selection is unidentified, propagate the resulting range into O_M and downstream values.
- **Shapley attribution:** enumerate all coalitions for small $|J|$; for larger $|J|$, use a declared permutation-sampling approximation and report uncertainty.
- **Partial identification:** optimize lower and upper bounds over $\Gamma_k(r)$, rather than filling missing values with means unless that imputation is explicitly modeled.

15. FAILURE CODES AND NON-NEGOTIABLE RULES

An implementation should use explicit failure states:

- **NI_INPUT:** required value not observed and no declared bound exists;
- **INFEASIBLE:** $\mathcal{F}^{SE} = \emptyset$;
- **NONBINDING:** mission declared but constitutive binding requirements fail;

- **UNIDENTIFIED_COUNTERFACTUAL:** x^0 , x^Ω , or attribution is not identified;
- **ZERO_CONFLICT_READOUT:** integrity denominator has no identifiable conflicts;
- **SILENT_LIFT:** a claim depends on undeclared augmentation;
- **SUPPORT_DEFEATED:** all minimal sufficient support bundles are undercut;
- **LEAKED_PREDICTION:** outcome information entered before prediction freeze;
- **PORTABILITY_NI:** ecosystem transfer cannot be evaluated from available evidence.

Two rules are absolute: **missing is not zero**, and a **model-generated counterfactual is never relabeled as an observed fact**.

16. DEVELOPMENT-STAGE STRESS TESTS

The architecture was iteratively pressure-tested against heterogeneous real enterprises and adversarial simulations. During development, one 45-enterprise distinct retrospective harness

Table 1: Minimum AI calculation contract. Every field carries a source / provenance identifier and an uncertainty or identification status.

Block	Required fields	Type / unit	Rule if missing or ambiguous
Time	t, T, δ	integer horizon; discount factor	Ask for horizon or run declared scenarios; never infer a calendar unit silently.
State	financial, capacity, trust, hazard, regime fields from Eq. (3)	currency, ratios, scores, categorical regime	Missing state is NI; scenario bounds may be used only if declared.
Topology	nodes, edges, governance rights	graph objects	Do not collapse a multi-entity system into one firm if control / resource edges are load-bearing.
Mission	loci b , constraints G_j , thresholds τ_j^{auth} , mission objective Ψ^M	Boolean / categorical functions and numerical thresholds	Mission statements without a computable rule are declaration-only; binding may remain 0 / NI.
Actions	candidate actions or continuous decision variables	action records with private value V , mission score, cash effects	If no action set exists, no optimization claim is allowed.
Opportunity	$\Omega^{\text{exo}}, \Omega^{\text{endo}}$	actions / options plus causal provenance	Mission attribution is NI unless a support path from mission architecture to opportunity is declared.
Binding	W_j , constitutive subset R_j^* , assurance a_j	binary components + $[0, 1]$ assurance	Missing constitutive requirement sets $q_j = 0$; missing assurance makes $\hat{\beta}_j = \text{NI}$.
Evidence	readouts, provenance DAG, support bundles	graph + source IDs + date / version	Failing E, A , or D downgrades the claim to NI / Open / qualified.
Uncertainty	scenarios, probabilities or uncertainty set	probability / scenario set	Never use chance and robust constraints simultaneously unless explicitly specified.
Policy support	instruments u_t , incentive rule, incidence matrix	cash + rule objects	Treat support as resource + incentive + institutional change, not cash only.

forced major additions including Payment / Customer Finance as a locus, receivables / inventory / debt state, audience-specific trust, organizational topology, stakeholder-vector outcomes, mission-core / implementation separation, endogenous ecosystem feedback, and market / capability-creation investment. A later heterogeneous 40-case battery, which was *not* disjoint from earlier development cases, was coded under a frozen v5 architecture as 30 PASS, 8 PRESSURE, 2 BOUNDARY, and 0 direct structural FAIL. These counts are finite-diagnostic, not accuracy estimates or population validation.

A quasi-blind pilot then froze directional prediction rules before outcome retrieval, recorded outcome leakage, and excluded leaked cases from scoring. Among adjudicable predictions, no observed disconfirmation was found in the pilot, while several outcomes remained NI. The point of this program is methodological: future validation should preregister coding rules, use held-out enterprises, employ independent coders, and define failure conditions before opening outcomes. The source package accompanying this preprint includes a case evidence ledger and a prediction-pilot ledger so that later revisions can preserve the development history rather than silently rewrite it.

17. COLLAPSE CONSISTENCY AND FALSIFIABILITY

MOCA must reduce to ordinary economic optimization when mission-specific architecture disappears. If

$$\Omega^M \rightarrow \emptyset, \quad G_j \rightarrow \text{always true}, \quad \Psi^M \rightarrow 0, \quad C^{\text{bind}} \rightarrow 0, \quad (84)$$

then

$$\mathcal{F}^{SE} \rightarrow \mathcal{F}^0, \quad x^{SE} \rightarrow x^0, \quad O_M, C_G, C_\Psi \rightarrow 0. \quad (85)$$

The framework is falsifiable at multiple levels. Examples include: mission loci cannot be coded with acceptable inter-rater reliability; binding architecture does not predict behavior under identifiable conflict; opportunity attribution cannot be separated from generic market growth; dynamic viability adds no predictive content beyond ordinary financial ratios; portability adds no information about replication failure; the readout layer never changes a decision or revision verdict; or held-out prospective predictions are systematically refuted.

18. LIMITATIONS

First, many terms are defined by counterfactual comparison and therefore require explicit benchmark assumptions. Second, enterprise-level private value and social value are not the same object. Third, mission opportunity can become an unfalsifiable catch-all unless mission-to-opportunity attribution is independently supported. Fourth, ecosystem state can become “everything else” unless the audit question fixes which institutional factors matter. Fifth, audience-specific trust vectors can become high-dimensional; only audiences causally relevant to the current decision should be included. Sixth, Bellman-style optimization does not imply that managers possess a true utility function; it is a disciplined representation of declared ranking and constraints. Seventh, the social-domain equations in this paper are not yet Coq-certified. The formal repository’s generic readout results provide a mathematical floor for

epistemic caution, not empirical proof of the organizational model.

19. CONCLUSION

MOCA converts a family of social-enterprise concepts into an executable management architecture. The central move is to stop treating mission as an organizational label and instead represent where mission creates opportunities, where it restricts choice, how it changes preference among admissible options, how those commitments are bound in governance and records, how they alter dynamic viability, and how strongly any resulting claim is identified by the available readouts. The framework preserves the original Dual Costs, TTD, Positional Governance, and Readout Condition as anchors while placing them in one calculation graph. An AI can therefore compute only what has been typed, benchmarked, and supported, can return NI where distinctions are unavailable, and can expose exactly which assumption or evidence node makes each result possible. That transparency is intended to be as important as the numerical answer.

ACKNOWLEDGEMENTS

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DISCLOSURE

An AI assistant was used for drafting, literature mapping, adversarial review of the architecture, and $\text{\LaTeX}$ production. The framing, the model, the choice of loci and benchmarks, the case coding, and the conclusions are the author's.

DATA AND CODE AVAILABILITY

The source bundle listed in Appendix D accompanies this preprint, including the machine-readable calculation contract, equation registry, knowledge-graph schema, worked example, case evidence ledger, and prediction-pilot ledger.

A. EQUATION INVENTORY FOR MACHINE PARSING

Table 2 lists stable identifiers for software and AI use. The identifiers, not the printed equation numbers, are the durable reference.

B. RECOMMENDED MISSION-LOCUS BINDING REQUIREMENTS

The following defaults are starting points, not natural laws. Projects should declare alternatives.

Locus	Candidate constitutive requirements
Allocation	rule, accounting record, distribution restriction, audit
Labour	eligibility rule, employment record, worker rights, voice / complaint mechanism
Sourcing	supplier rule, traceability record, supplier rights / voice where mission-relevant, audit
Product	price / quality / access threshold, beneficiary definition, transaction record
Payment	financing rule, fee / interest ceiling, receivables record, affordability test
Transaction	non-completion objective, protected-party rights, separate condition record, conflicts disclosure, audit, admissible capital
Data	collection / use rule, consent / rights, access log, audit, model documentation
Capital	control-right lock, return / horizon rule, transfer / exit restriction, disclosure
Lifecycle	service / take-back / warranty rule, obligation record, funding reserve, completion audit

C. REAL-CASE BATTERY SUMMARY

The detailed source URLs and case-by-case mechanism coding are supplied in `moca_case_evidence_ledger.csv`. The 40-case frozen-v5 battery includes Grameen Shakti, KickStart International, Revolution Foods, Everytable, Rebel Ventures, Eat My Lunch, Al Sanabel, d.light, M-KOPA, Peepoople, Clean Team Ghana, Jibu, Solar Sister, Bridge International Academies, Hapinoy, Green Monday, KOTO, Angaza, Siam Organic / Jasberry, Waste Concern, Doi Tung Development Project, Akha Ama Coffee, Dairy Home, Liberation Foods, Specialisterne, TerraCycle, Better World Books, Ben & Jerry's, TOMS, Ecofiltro, Riders for Health, SEKEM, The Big Issue, Aarong, Delancey Street Foundation, Compartamos Banco, SKS Microfinance, Thistle Farms, BRAC / Aarong Dairy, and Grameen Bank. The coding purpose was structural pressure testing, not ranking these organizations.

D. SOURCE-PACKAGE FILES

The source bundle is intended to be self-contained:

- `moca_technical_paper.tex`: this paper;
- `moca_refs.bib`: bibliography;
- `moca_ai_contract.json`: machine-readable input / output contract;
- `moca_equations.json`: equation registry with inputs, outputs, tiers, and missing-value rules;
- `moca_kg.jsonld`: knowledge-graph schema;
- `moca_example_input.json` and `moca_example_output.json`: executable worked example;
- `moca_case_evidence_ledger.csv`: 40-case stress-test ledger;
- `moca_prediction_pilot.csv`: prediction-freeze pilot ledger.

Table 2: Core equation inventory. IDs are stable names for software and AI use.

ID	Output	Calculation class	Primary inputs
E-READOUT	$R_{a,t}$	direct / readout	latent-model interface, source transform, errors
E-STATE	X_t, S_t	assembly	financial, capacity, trust, hazard, ecosystem, topology
E-BIND	$q_j, \hat{\beta}_j$	Boolean + assurance	constitutive binding components, assurance
E-OMEGA	Ω^M	union / fixed point	exogenous options, mission, binding, state, ecosystem
E-FEAS	$\mathcal{F}^{SE}$	filtering / optimization	actions, constraints, authorized thresholds
E-ECON	O_M, C_G, C_Ψ, C_M	four benchmark optimizations	$V, \Psi^M, \mathcal{F}^0, \Omega^M, G_j$
E-SHAPLEY	ϕ_j	coalition attribution	coalition value function $C(S)$
E-CASH	Liq_{t+1}	accounting	operating cash, support, bind cost, investment, debt, working capital
E-TTD	revenue / gap / hazard / trust	dynamic plug-in	TTD parameters and state
E-OBLIG	Q_{t+1}^M	stock–flow	new obligations, fulfilment, decay
E-DYN	W^{SE}, S_{t+1}	dynamic optimization	policy, transition, horizon, uncertainty
E-INTEGRITY	I_j	conditional ratio	identifiable conflict events and honoring decisions
E-DRIFT	$\text{Drift}_{j,t}$	rule comparison	observed vs. authorized threshold, conflict
E-INCIDENCE	$\Delta V_g, Y^{\text{soc}}$	distributional	stakeholder-specific opportunities, costs, outcomes
E-READOUTAUDIT	$\mathcal{I}_q, \text{RC}(d)$	set identification / provenance	readouts, assumptions, support bundles
E-FALSIFY	hit / refute / NI	prediction audit	frozen prediction, leak flag, outcome support

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